

Per-Stream Terminal Valence Closure:

A Boundary Hypothesis and Framework for Empirical Testing

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Abstract

Could the end of a conscious life reveal a mathematical constraint on the experiences that preceded it? Per-Stream Terminal Valence Closure (PTVC) makes that question a precise, falsifiable candidate boundary hypothesis for affective science, consciousness research, measurement theory and stochastic processes. On its primary finite signed-measure branch, it asks whether each separately routed maximal qualifying conscious stream carries exactly equal positive and negative Jordan masses across its complete strict-pre-endpoint domain, so that the terminal signed residual is zero. The equality is intrastream and nontransferable. This compact boundary condition is compatible with the full diversity of lived experience: lives may differ in gross valence variation, duration, intensity, order, agency, and structure.

$$Z_i^\star := \nu_i(\Omega_i) = V_i^+ - V_i^- = L_i^{\mathrm{phen}}(T_i^\star -)$$

$$\mathrm{PTVC}_i \Leftrightarrow V_i^+ = V_i^- < \infty \Leftrightarrow Z_i^\star = 0$$

The boundary condition gains scientific force when representation, neutral origin, exposure, stream ownership, endpoint, observation and recovery rules are fixed before outcome inspection. One sharp consequence concerns endpoint timing. Under a declared reference law, let the endpoint be independent of the entire jointly measurable pre-endpoint ledger path, with a timing density positive almost everywhere on an interval. If exact closure holds at that endpoint almost surely, the ledger must equal zero at almost every eligible time on almost every eligible path. This obstruction gives independent, continuously distributed ending models a precise test. State-coupled timing, first hitting, absorption, endpoint atoms and singular carriers offer alternative models with different testable signatures.

PTVC offers a research program built for prospective evaluation. The first steps are independently calibrated measurements and model recovery. Random-cut comparisons, endpoint-surprise tests, completion-potential transport and concealed-outlet experiments then probe distinct consequences under frozen assumptions. A strong candidate should outperform equally informed rivals on held-out data and invite independent replication. Evidence for one proposed mechanism and evidence for universal terminal equality remain distinct: a precise behavioral signature can favor a route while the primary boundary continues to have its own scientific tests. The aim is cumulative discovery, with each experiment providing an interpretable result and a stronger next question. Together, these investigations ask whether conscious experience obeys a terminal organizing principle.

Section 14 develops a distinctive empirical proposal: mosquitoes assigned to different adult histories encounter a concealed, randomly designated continuing-life outlet, while a separate test assigns the outcome only after commitment. Different history durations, feeding delays and developmental outcomes create additional prospectively declared contrasts. When the assignment is genuinely uniform and independent of recorded choice, the exact matching baseline is 1/N.

Keywords: affective valence; pleasure and pain; phenomenal consciousness; animal consciousness; measurement theory; signed measures; partial identification; Per-Stream Terminal Valence Closure (PTVC); Law of Fairness

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1. Scientific question, scope, and research opportunity

A balance rises linearly from zero to one in an hour, then returns to zero at the same rate. At a uniformly chosen instant it is positive with probability one. What singles out the moment a conscious stream ends?

A boundary hypothesis with a precise domain

PTVC asks whether each qualifying conscious stream ends with exactly equal totals of positive and negative felt experience. The balance can cross zero earlier. Compensation may span many events without conscious tracking or intent.

Contribution and novelty

Benjamin Franklin (1725) anticipated the central intuition of the Law of Fairness when he argued that "every Creature hath as much Pleasure as Pain." Nietzsche (1910, §12) asked whether "pleasure and pain should be so closely connected" that the greatest pleasure might require the greatest pain. Kahneman et al. (1997) showed when integrating momentary experienced utility ranks episodes. On one scale, two streams can admit zero at incompatible offsets (Section 4). Section 6 links time off balance to closure under independent endpoint timing. Section 10 tests forecasts in a reward task, films and diaries.

The Science of Fairness models a choice between death and continued life and proposes nonterminal and developmental comparisons (Takowsky, 2026, Sections 19.2, 23.4.4, and 29.4.5). Here we propose a terminal mosquito experiment with a separate arm in which assignment follows outlet choice.

2. Formal object and primary finite-measure claim

For a separately routed maximal qualifying stream i, fix a signed phenomenal representation. Its nonempty measurable truth domain (Ω_i, Σ_i) covers the entire stream strictly before its endpoint. On the primary branch, ν_i is a finite real-valued signed measure on Σ_i with Jordan decomposition $\nu_i = \nu_i^+ - \nu_i^-$ into mutually singular positive measures. V_i^+ and V_i^- are their Jordan masses. Their difference defines the terminal signed residual independently of PTVC. The hypothesis requires every qualifying stream to admit this finite representation with equal masses.

Assign each owned contribution a time through a measurable map to the stream's clock. The ledger at a time is the signed mass of contributions through that time. Its left limit counts those strictly earlier. Every time in the primary domain precedes the endpoint, so continuity from below of the finite signed measure gives the full signed mass in the terminal left limit. For an infinite endpoint, take this limit along increasing finite times.

Equation (2.1). Primary finite signed-measure branch

$$\nu_i = \nu_i^+ - \nu_i^-, \nu_i^+ \perp \nu_i^-$$

$$V_i^{\pm} := \nu_i^{\pm}(\Omega_i), V_i^{\text{gross}} := V_i^+ + V_i^- = |\nu_i|(\Omega_i)$$

$$Z_i^\star := \nu_i(\Omega_i) = V_i^+ - V_i^- = L_i^{\text{phen}}(T_i^\star -)$$

$$\text{PTVC}_i \Leftrightarrow V_i^+ = V_i^- < \infty \Leftrightarrow Z_i^\star = 0$$

Within a stream, $V_i^+ = V_i^- > 0$ gives exact closure with positive gross variation. Two separate atoms of mass plus one and minus

one at the same instant contribute two units of gross variation on the original domain. Their net increment on the clock is zero.

A latent ledger requires an exposure model

For the time-density branch in Equation (2.2), assume that signed mass on the clock has an integrable density with respect to the clock's sigma-finite exposure measure. For a whole stream, S_i^{θ} is the truth-domain origin and $L_i^{\text{phen}}(S_i^{\theta}-)=0$. For later entry, see Equation (4.2). On the primary convention, $g_{\theta,i}$ is a dimensionless time-chart, unity, or ownership factor and q_i^{exp} has exposure-per-time units; on the unity-clock alternative, $q_i^{\text{exp}}\equiv 1$ and $g_{\theta,i}$ carries exposure-per-time units. Their product is the density of the clock push-forward of the nonnegative experienced-exposure measure μ_i^{phen}. Then v_i^{real} denotes this signed density, the realized neutral-relative phenomenal value per unit of exposure.

Equation (2.2). Exposure-weighted latent ledger

$$L_i^{\text{phen}}(t)=L_i^{\text{phen}}\left(S_i^{\theta}-\right)+\int_{S_i^{\theta}}^{t} g_{\theta,i}(s)\,q_i^{\text{exp}}(s)\,v_i^{\text{real}}(s)\,ds$$

Bipolar and generalized branches

The net signed measure need not exhaust phenomenal structure. On a bipolar branch, let β^+ and β^- be finite nonnegative component measures on the same domain, with $\nu = \beta^+ - \beta^-$. The minimality of the Jordan decomposition gives a unique nonnegative common measure ζ such that

Equation (2.3). Bipolar decomposition and gross magnitude

$$\beta^+ = \nu^+ + \zeta \, , \beta^- = \nu^- + \zeta \, .$$

Consequently, primitive gross magnitude and net total variation satisfy

$$\beta^+(\Omega) + \beta^-(\Omega) = |\nu|(\Omega) + 2\,\zeta(\Omega).$$

Each unit of mass common to both components adds two units to their combined magnitude and cancels in their difference (Folland, 1999). Generalized Terminal Neutrality (GTN) remains a separate nonadditive or functional-zero branch unless a valid bridge connects it to the additive or bipolar terminal object. Vector, ordered and set-valued representations remain open branches.

Notation at a glance

Measures (Section 2). $\nu_i, \nu_i^{\pm}$ are the signed measure and Jordan components; $V_i^{\pm}, V_i^{\mathrm{gross}}$ are their masses and total variation.

Ledger and time (Section 2). $L_i^{\mathrm{phen}}(t), Z_i^{\star}$ denote the ledger and terminal residual; $S_i^{\theta}, T_i^{\star}$ denote stream origin and endpoint. A trailing minus indicates a left limit.

Exposure (Section 2). $g_{\theta,i} q_i^{\mathrm{exp}}$ is experienced-exposure density; v_i^{real} is value per unit exposure.

Representation (Section 4). a_i, b_i are the neutral-origin shift and positive scale.

Observation (Section 5). $K_{i,\eta}^{\mathrm{obs}}$ is the observation kernel; $P_i^{Z}, P_{i,\eta}^{\mathrm{obs}}$ denote the latent residual and observed laws, respectively.

Probability (Sections 6, 8–9). P_i^0 is a declared reference law; $P^{\pi,g}$ is a policy-and-regime law. $\tilde{L}_i, \mathcal{L}$ denote ledger paths in Section 6.

Residual sets. Definitions and guarantees: Section 5; reporting fields: Section 13.

Contextual reuse. L is the fixed-cut prefix in Section 8; h denotes history in Sections 7–8, while $h_{i,\varepsilon}$ is the completion potential in Section 9. X denotes retained state in Section 8 and the partition vector in Section 9.

3. Quantifiers, ownership, routing, and falsification

Stream-level adjudication of the universal claim

Under [SET-ALL], every separately routed maximal qualifying stream in the domain fixed by the protocol satisfies exact closure. Section 5 gives the residual set criterion for a counterexample from one qualifying stream. A probability-one claim allows exceptions confined to a set of probability zero under the specified random-stream law.

Bearer and routing rules are causal, not clerical

Comparative consciousness research supports testing stream individuation and eligibility across species, including nonverbal organisms (Birch et al., 2020). Routing must be exhaustive, nonduplicative, time-consistent and outcome-independent. For example, two periods of wakefulness separated by uncertain consciousness may belong to one stream or two. Compute the residuals under each admissible routing. A counterexample survives the ambiguity if every routing forces at least one qualifying stream off zero. Keep separate nonnegative accounts for unresolved positive and negative contributions so their slacks cannot cancel. Section 13 gives the conditions for generalizing from the observed streams.

Modal claims form a partly ordered family

Distinguish existence of a completing continuation, positive completion probability, probability one under a fixed policy, and closure on every admissible continuation. For [SUPPORT],

specify the space, topology and target. For a real residual in its usual topology, a closed target contains the support exactly when it has probability one. A probability-one path-space event can omit support points if the event is not closed. Almost-sure attainability requires one admissible policy attaining probability one. Limit-sure attainability permits probabilities arbitrarily close to one without requiring attainment (Asadi et al., 2025; Chatterjee et al., 2016). All-policy and all-intervention claims require a structural model and an admissible counterfactual domain. Extending a fixed-policy result to either scope or [SET-ALL] requires a separate bridge.

Target and route are separate axes

Evidence for a mechanism supports closure wherever the verified mechanism entails it. Section 13 records the boundary and mechanism verdicts separately.

4. Representation, neutral origin, and delayed entry

Neutral origin must be calibrated independently

A scalar ledger requires a sign convention, cross-sign comparability, an exposure unit, and a neutral origin. Under the admissible affine recoding $v'=a_i+b_iv$ with $b_i>0$, a stream-constant positive scale preserves a zero residual, whereas on positive-exposure records an additive shift preserves zero only when $a_i=0$. Specify the origin class, its basis, complexity and envelope before examining terminal totals. Repeat independently justified neutral anchors across contexts in a separate sample. Across 16 volunteers, the same moderate heat drew an unpleasant mean sensation rating against painless warmth and a pleasant one against intense pain (Leknes et al., 2013).

Equation (4.1). Origin sensitivity under affine transformation

$$\widetilde{Z}_i^{\star} = a_i E_i^{\mathrm{tot}} + b_i Z_i^{\star}, b_i > 0, E_i^{\mathrm{tot}} = \mu_i^{\mathrm{phen}}(\Omega_i) < \infty$$

If total exposure is positive, choosing $a_i = -b_i Z_i^{\star}/E_i^{\mathrm{tot}}$ forces the transformed residual to zero. A review of patient-reported outcomes detected response shifts in 793 of 4,868 effects investigated (Sawatzky et al., 2024). Perturb independently validated positive and negative channels to estimate their exchange and test state dependence and hysteresis. At a fixed origin, six positive and four unsigned negative units require 1.5 positive units per negative unit. Exchange rates from 0.9 to 1.1 leave residuals from 1.6 to 2.4. Lim et al. (2009) used magnitude estimation to derive bipolar semantic anchors. The resulting

hedonic scale and magnitude estimates agreed closely for 26 food names.

Study entry and stream origin are distinct

When observation starts late or ends early, whole-stream inference must account jointly for the prefix, observed interval and suffix on the same latent scale.

Equation (4.2). Delayed-entry decomposition

$$Z_{i,\mathrm{pre}}(a) := L_i^{\mathrm{phen}}(a-)$$

$$Z_{i,\mathrm{obs}}([a,b)) := L_i^{\mathrm{phen}}(b-) - L_i^{\mathrm{phen}}(a-)$$

$$Z_{i,\mathrm{suf}}(b) := L_i^{\mathrm{phen}}\left(T_i^{\star}-\right) - L_i^{\mathrm{phen}}(b-)$$

$$Z_i^{\star} = Z_{i,\mathrm{pre}}(a) + Z_{i,\mathrm{obs}}([a,b)) + Z_{i,\mathrm{suf}}(b)$$

Allow any real prefix, an observed contribution equal to one minus that prefix, and any nonnegative suffix. Their possible totals fill $[1,\infty)$ despite the unbounded components.

Joint calibration: a sharp three-interval example

For three signed interval contributions, suppose the jointly valid restrictions are

$$x_1 + x_2 \in [0.7,0.9], x_2 + x_3 \in [0.3,0.5],$$

$$x_2 \in [-0.2,0.2].$$

Writing $J = x_1 + x_2 + x_3$ gives

Equation (4.3). Sharp residual range from joint calibration

$$J = (x_1 + x_2) + (x_2 + x_3) - x_2 \in [0.8, 1.6].$$

The feasible vectors $(0.5, 0.2, 0.1)$ and $(1.1, -0.2, 0.7)$ attain the endpoints. Their convex combinations satisfy the same linear constraints and attain every intervening total. Without the bound on x_2, the two observed sums can remain fixed while J becomes arbitrarily positive or negative.

For a complete qualifying stream, the calibrated range rules out closure. For a fragment, it identifies the contribution’s sign. At unit scale and four exposure units, an origin error of at most 0.10 per unit shifts the total by at most 0.40. The resulting outer interval [0.4, 2.0] excludes zero. An allowance of 0.20 first admits zero to the outer interval.

Suppose independent calibration gives each instrument total as the true residual plus a common offset times exposure. Totals in [0.8, 1.6] and [1.0, 1.2] at exposures of four and two require closing offsets in [0.20, 0.40] and [0.50, 0.60]. Their 0.10 gap forces an absolute residual rate of at least 0.05 in one stream. The largest count of ranges sharing an admissible offset bounds how many streams can close together. For positive rates (1, 2, 4) and negative magnitudes (1, 2, 3), weights (1, −2, 1) cancel the common exchange rate and offset, leaving one. Their absolute sum is four, so the largest absolute residual rate is at least 0.25. Errors bounded by 0.05 per rate equation can shift the contrast by at most 0.20, lowering that bound to 0.20.

Factorization, scalarity, and gauge freedom

On the density branch, reciprocal positive predictable rescaling preserves the exposure–intensity product. In independent calibration data, perturb one factor while measuring or fixing the others and the observation channel. Emotional pictures shortened judged durations of later visual or auditory stimuli (Lui et al., 2011). Record physical and judged duration across affect and attention conditions. For scales from ratings, ranks, embeddings, factors or logits, ask which stimulus pair differs more in felt intensity. Fit spacings to calibration judgments (Maloney & Yang, 2003) and predict held-out comparisons. Across six emotions, 71–97 percent of participants made intensity comparisons consistent with the difference-scaling axiom. Among them, an average of 64 percent gave direct ratings that departed from the inferred interval scale (Junge & Reisenzein, 2016). Report test power (Reisenzein & Junge, 2024). Test the cross-sign exchange against additive-conjoint independence or separability, cancellation, solvability and held-out comparisons (Luce & Tukey, 1964). After graduation, 50 percent of 66 students reported happiness and sadness together, versus 20 percent of 49 on a typical day (Larsen et al., 2001). Paired ratings can distinguish mixed and neutral reports with the same net score (Cacioppo & Berntson, 1994). Measuring overlap in Equation (2.3) requires both magnitudes in common calibrated units.

5. Observation, exactness, and the inference ladder

On the scalar branch, let P_i^Z denote the probability law of the latent terminal residual on the real line. For each application, declare what varies under this law: a model ensemble conditional on stream information, a random-stream population, or another specified sampling scheme. State the exchangeability or structural assumptions connecting available records to that law; record posterior uncertainty about a realized residual as a separate evidence object. Observed reports, physiology, behavior, neural signals, and clinical records arise through a Markov kernel $K_{i,\eta}^{obs}$ from that latent space to a specified observation space; η indexes measurement conditions such as resolution, care context and retention (Section 12). The observed law $P_{i,\eta}^{obs}$ is the kernel push-forward of P_i^Z. Measurement can therefore narrow the spread of the readings even when the underlying residuals are just as variable. For longitudinal observations, incorporate and validate the conditional law of histories given the residual, or enlarge the latent state.

Equation (5.1). Observation-kernel push-forward

$$P_{i,\eta}^{\mathrm{obs}}(A)=\int K_{i,\eta}^{\mathrm{obs}}(A\mid z)\,P_i^Z(dz)$$

Target geometry and exactness levels

For a nonempty outer set guaranteed to contain the stream's true residual under the maintained assumptions, a singleton at zero establishes exact closure. A set that excludes zero supplies a counterexample to universal closure. Admitting both zero and nonzero values leaves exact closure unresolved.

For a finite-band target, the residual set is band-equivalent when wholly inside the acceptance set, band-adverse when disjoint, and band-indeterminate when it overlaps both acceptance and nonacceptance. The interval $[-\varepsilon, \varepsilon]$ is a scalar special case; a law-level band-probability claim separately specifies its stream or population law, minimum mass threshold, and error rule. An exact zero atom is positive mass at zero itself.

Inference must preserve the object type

The observable population law and the maintained restrictions determine the identified set (Manski, 2003). An estimated bound has sampling uncertainty. State whether a confidence procedure covers the true parameter or the entire identified set (Imbens & Manski, 2004). A posterior region depends on its prior and model. A sensitivity region describes the restrictions varied.

Equivalence requires affirmative evidence

Suppose the truth-containing range [0.8, 1.6] in Section 4 is tested against an independently specified tolerance of two units. It excludes exact zero and lies wholly within that tolerance (Schuirmann, 1987; Lakens, 2017). If those same bounds instead project a 95% joint confidence set, the conclusions carry the procedure's repeated-sampling coverage.

Exact atoms require stable inverse recovery

Population identification and stable finite-sample atom recovery are distinct requirements. When the observation kernel is

continuous at zero in total variation, atomless latent laws concentrating near zero can approximate the observed law of a zero atom arbitrarily closely. For independent observed records, uniform finite-sample exact-atom discrimination over that unrestricted class therefore requires a protocol-fixed separation or structural restriction. In a constructed example, 100 independent readings have known centered Gaussian measurement noise. A 5% upper-tail test uses their sum of squares to test a zero residual. For an independent centered Gaussian residual, power reaches 80% at a residual-to-noise standard deviation ratio of about 0.64. At a ratio of 0.10, power is 5.8%.

6. Random endpoints: a sharp structural discriminator

Independent positive-density endpoint proposition

Let E_i be a prospectively declared positive-probability event on which the stream, frozen routing and truth domain, finite ledger, and pre-killing extension exist on a deterministic interval I; write $P_i^0=P(\cdot|E_i)$. Under this same law, let $\tilde{L}_i(t-)$ be jointly measurable with respect to the product of its path-generated sigma-field $\mathcal{F}_i^L=\sigma\{\tilde{L}_i(s-):s\in I\}$ and the Borel sigma-field on I. Let T_i^{exo} be independent of $\mathcal{F}_i^L$. Assume T_i^{exo} is I-valued, has a Lebesgue density positive almost everywhere on I, and $P_i^0\{\tilde{L}_i(T_i^{exo}-)=0\}=1$.

Equation (6.1). Independent positive-density endpoint obstruction

$$0=P_i^0\left(\tilde{L}_i\left(T_i^{\mathrm{exo}}-\right)\neq 0\right)=\int_I P_i^0\left(\tilde{L}_i(t-)\neq 0\right) f_{T_i^{\mathrm{exo}}}(t)\,dt$$

The integrand is nonnegative. Tonelli's theorem and the positive density give the product-measure statement: $P_i^0\{\tilde{L}_i(t-)\neq 0\}=0$ for Lebesgue-almost every $t\in I$; Fubini's section conclusion equivalently gives $\tilde{L}_i(t-)=0$ for Lebesgue-almost every t on a P_i^0-probability-one set of paths. If paths are left-continuous P_i^0-almost surely, equality holds at every interior time. The left endpoint requires right continuity or a separate value condition. The incoming cumulative ledger in Section 2 is left-continuous. On an extension of that form, subtracting its values at two interior times gives zero net mass from the first time up to, but excluding, the second. A calibrated nonzero

increment with positive probability on the same eligible law would rule out independence and closure together.

What the discriminator reveals

For the two-hour path in Section 1, an independent uniform ending has a one-in-ten chance of finding the balance within 0.1 unit of zero. The favorable times occupy six minutes at each end. Exact zero occupies none of the interval’s duration. The obstruction holds at every fixed level. The independent calibration in Section 4 selects zero.

Empirical use

Compare disclosed, concealed and unexpectedly rapid horizons, recording what the participant knew. An anticipated ending can change motivation and emotional priorities, making advance planning a rival explanation (Carstensen et al., 1999; Carstensen, 2006). An ending can surprise the participant and still depend on the ledger path.

Approximate dependence yields a graded pressure test

Prespecify which covariates enter the conditional laws for eligible streams. Choose a nonnegative tolerance, a jointly measurable pre-endpoint ledger path and a Borel subset of the interval with finite positive Lebesgue measure. Set

$$X_i := \left(\tilde{L}_i(t-)\right)_{t\in I},\ \varepsilon \geq 0,\ J \subseteq I,\ 0<\lambda(J)<\infty,$$

$$g_{i,\varepsilon}(t,c) := P_i^0\left(|\tilde{L}_i(t-)| > \varepsilon \mid C_i = c\right),$$

$$p_{i,\varepsilon} := P_i^0\left(\left|\tilde{L}_i\left(T_i^{\text{probe}}-\right)\right| > \varepsilon\right),$$

$$\rho_i := \mathbb{E}_{P_i^0}\left[d_{\text{TV}}\left(P_{X_i,T_i^{\text{probe}}\mid C_i}, P_{X_i\mid C_i} \otimes P_{T_i^{\text{probe}}\mid C_i}\right)\right].$$

The total-variation distance is defined by

$$d_{\text{TV}}(P,Q) := \sup_A |P(A) - Q(A)|.$$

Assume standard-Borel path and covariate spaces, measurable evaluation, and the required regular conditional laws. Suppose that

$$f_{T_i^{\text{probe}}\mid C_i}(t \mid C_i) \geq m_J > 0.$$

The density bound holds for Lebesgue-almost every time in the declared set and almost surely under the eligible-stream reference law.

Equation (6.2). Approximate endpoint-dependence occupation bound

$$\mathbb{E}_{P_i^0}\left[\int_J g_{i,\varepsilon}(t, C_i)\, dt\right] \leq \frac{p_{i,\varepsilon} + \rho_i}{m_J}, m_J > 0$$

After averaging over covariates, the off-band probabilities under the joint and conditional-product laws differ by at most ρ_i. The density floor under the conditional-product law then gives Equation (6.2). In a constructed example, suppose expected off-band occupation in the analysis set is at least three time units, the conditional probe-density floor there is 0.10 per unit, and the mean conditional total-variation distance is at most 0.05. The

probe then lands outside the band with probability at least 0.25. The 0.05 ceiling concerns dependence on the full extended path. If observations stop at the endpoint, justify that ceiling through the probe design or an extension model.

Information required by endpoint enrichment

Under the eligible-stream law of Equation (6.2), compare two ways of pairing an endpoint with a path at a given covariate value. Draw them jointly, or draw them independently from their respective conditional marginals. In both cases, record whether the path lies within the band just before the paired endpoint on the common evaluation domain. Logarithms are natural.

$$\begin{aligned} P_c &:= \mathrm{Law}\big((T,\mathcal{L}) \mid C=c\big), \\ Q_c &:= P_{T\mid c} \otimes P_{\mathcal{L}\mid c}, \\ A_\varepsilon &:= \{\,|\mathcal{L}(T-)| \le \varepsilon\,\}, \\ p_c &:= P_c(A_\varepsilon), q_c := Q_c(A_\varepsilon), \\ p &:= \mathbb{E}\, p_C, q := \mathbb{E}\, q_C \end{aligned}$$

Equation (6.3). Information bound for endpoint enrichment

$$I(T;\mathcal{L} \mid C) \ge \mathbb{E}\, \mathrm{d}_{\mathrm{Ber}}(p_C \,\|\, q_C) \ge \mathrm{d}_{\mathrm{Ber}}(p \,\|\, q)$$

$$\mathrm{d}_{\mathrm{Ber}}(p \,\|\, q) = p \log \frac{p}{q} + (1-p) \log \frac{1-p}{1-q}$$

Relative entropy contracts under event recording, and joint convexity gives the second inequality (Cover & Thomas, 2006). Standard extended-value conventions apply: for p = 1 and 0 < q < 1 the lower bound is log(1/q), while for q = 0 < p it is infinite.

7. Reachability, endpoint support, and solvency

A residual must be cancelable by an admissible future

For a realized stream-owned prefix h_t and prospective endpoint u, define the admissible continuation class $C_i(t,u|h_t)$ from the information available at that prefix. For a nonempty class, let Cap_i^+ be the supremum of the positive parts of attainable net increments and Cap_i^- the supremum of their negative parts, expressed as nonnegative magnitudes. Either capacity is zero if no increment points in that direction. A closing continuation must contribute the negative of the current ledger:

Equation (7.1). Directional reachability necessity

$$-\mathrm{Cap}_i^+(t,u\mid h_t)\le L_i^{\mathrm{phen}}(t)\le \mathrm{Cap}_i^-(t,u\mid h_t).$$

On the time-density branch, a deficit of twenty units takes at least two hours to offset at a maximum positive rate of ten units per hour. Capacity alone may still be insufficient. If the only attainable increments are minus two and plus two, neither cancels a prefix of one. Compactness can secure the extremes while leaving such gaps (Aubin, 1991). An attainable interval containing the required increment gives a completion.

Endpoint support localizes the required structure

Let $T > 0$ be a stopping-time endpoint, possibly infinite, and let $N_t = 1_{\{T \le t\}}$ have predictable compensator A_t. Assume that $L(t-)$ is predictable in the same filtration and defined on the needed at-risk domain. For a finite horizon u and $\delta \ge 0$, the compensator identity gives

Equation (7.2). Endpoint event and predictable compensator

$$\Pr\{T \le u, |L(T-)| > \delta\}$$

$$= \mathbb{E} \int_{(0,u]} 1_{\{|L(t-)| > \delta\}} \, d A_t .$$

The integral against $d N_t$ is the indicator of the endpoint event in Equation (7.2); replacing $d N_t$ by its compensator inside expectation proves the identity (Andersen et al., 1993; Protter, 2005). For exact incoming closure, the right side at $\delta = 0$ must vanish. On every finite horizon, compensator mass is therefore confined to incoming zeros almost surely. For $d A_t = Y_t \lambda_t \, d t$, with $Y_t = 1_{\{T \ge t\}}$, the probability of ending at an incoming zero by u equals the expected integrated intensity over the at-risk times at zero. If those times have zero duration almost surely, positive endpoint probability there requires a singular compensator component in this filtration. Projection to an observed filtration requires a joint observation model for the hazard and ledger-band probability.

Endpoint adjudication is versioned evidence

Candidate crossing, Estimated endpoint, Confirmed under protocol, Revised endpoint, or No adequate estimate are distinct statuses. Dwell time, return logic, competing events, reliability, and revision rules are frozen before ledger inspection. Recovery or reclassification preserves the original analysis version and propagates endpoint uncertainty. Functional MRI or EEG detected task responses in 60 of 241 participants without an observable behavioral response to commands and in 43 of 112 who responded behaviorally (Bodien et al., 2024). Adjudicate the end of qualifying experience separately using serial multimethod findings. Record its uncertainty alongside last report, last response and biological death (Casali et al., 2013; Parnia et al., 2023; Manasova et al., 2026).

Endpoint branches are empirical alternatives

Test occupation on each singular carrier. On a first-hitting branch with a later ending, the first zero eligible under the entry or return rule must coincide with the independently adjudicated endpoint. A jump to zero at the endpoint can leave a nonzero incoming balance because the primary ledger counts only earlier contributions. On an inclusive branch, death-associated mass requires phenomenal evidence and boundary ownership. Biological-death admissibility further requires the relevant stream’s terminal closure by death. The model permits experience to continue at zero or to end while biological functions continue under life support.

8. Optional mechanism routes and path-class constraints

Finite variation restricts exact diffusion bridges

A finite signed-measure ledger has finite variation; a continuous version therefore has zero continuous quadratic variation. Suppose an exact path bridge identifies it with a reserve-coupled semimartingale whose continuous local-martingale coefficient is the product of a positive ledger–reserve conversion constant, the remaining reserve, and the diffusion coefficient, while the remaining terms have finite variation. Writing continuous quadratic variation in Equation (8.1), exact path equality requires both increments to vanish:

Equation (8.1). Finite-variation obstruction to an exact diffusion bridge

$$\left[L_i^{\mathrm{phen}}\right]_t^c - \left[L_i^{\mathrm{phen}}\right]_{S_i^\theta}^c = 0\,,$$

$$\left[L_i^{\mathrm{rc}}\right]_t^c - \left[L_i^{\mathrm{rc}}\right]_{S_i^\theta}^c = c_{\mathrm{LU}}^2 \int_{S_i^\theta}^{t} U_i(s)^2\, \sigma_i(s)^2\, d\,s = 0\,.$$

Assume the product of reserve and diffusion coefficient is predictable and locally square-integrable in time almost surely. Exact path equality then forces their product to vanish for Lebesgue-almost every time, almost surely; wherever reserve remains positive, the diffusion coefficient must vanish in the same sense. A nondegenerate diffusion remains available as an approximate, terminal-only, or consequence bridge (Protter, 2005). Jumps require separate quadratic-variation accounting.

Reserve and endpoint require a tested bridge

Construct the reserve proxy without terminal-neutrality labels. Test whether it predicts beyond prognosis, frailty, disease severity, treatment, care setting, device dropout and observation availability, and whether it transports across sites and perturbations. A separately tested bridge must connect reserve exhaustion to the end of the qualifying stream.

Mechanism-localization discipline

An intervention may change assignment, information, policy, action, physical transition, outcome, retention, endpoint, routing, or observation. The earliest distinguishable change in a conditional law is a candidate causal locus only under the declared identification, timing, measurement-lag, and observation assumptions. In continuous time, causal weighting additionally requires positivity and an appropriate local-independence structure (Røysland, 2011).

Action-mediated route

Under the protocol-fixed action-first factorization, let h be the predecision history, g the declared continuation regime, and $\mathcal{A}^{\mathrm{do}}(h)$ the measurable admissible action domain. Let π be the fixed policy and p^{comp}(a | h, g) the measurable probability that the complete continuation closes after action a under that regime. F_{comp} denotes the induced joint action–suffix completion event. With all policy mass on $\mathcal{A}^{\mathrm{do}}(h)$, averaging p^{comp} over that distribution gives Equation (8.2). Equation (8.3) reweights the actions while keeping the continuation law after each action fixed. Using a law estimated under another policy requires support for the evaluated actions and a causal justification for transporting the conditional law (Section 14).

$$p^{\mathrm{comp}}(a \mid h, g) := P_g\left(F_{\mathrm{comp}} \mid h, a\right)$$

$$\pi\left(\mathcal{A}^{\mathrm{do}}(h) \mid h, g\right) = 1$$

Equation (8.2). Fixed-policy probability-one completion

$$P^{\pi, g}\left(F_{\mathrm{comp}} \mid h\right) = \int_{\mathcal{A}^{\mathrm{do}}(h)} p^{\mathrm{comp}}(a \mid h, g)\, \pi(d\, a \mid h, g)$$

$$P^{\pi, g}\left(F_{\mathrm{comp}} \mid h\right) = 1$$

$$\Leftrightarrow p^{\mathrm{comp}}(a \mid h, g) = 1 \; \pi(\cdot \mid h, g)\text{-almost everywhere}$$

Finite soft guidance is weaker than exact selection. Write $q(a)=p^{\mathrm{comp}}(a\mid h,g)$ and fix a baseline policy π_0, a finite exponent $\beta>0$, and $Z_\beta=\int q(a)^\beta\,\pi_0(da)>0$. Reweighting by q^β yields

Equation (8.3). Finite soft-guidance completion probability

$$\pi_\beta(da)=Z_\beta^{-1}\,q(a)^\beta\,\pi_0(da),\quad P^{\pi_\beta,g}(F_{\mathrm{comp}}\mid h)=\frac{Z_{\beta+1}}{Z_\beta}.$$

The ratio equals one exactly when π_0 assigns no mass to $0<q<1$; positivity of Z_β then requires mass at $q=1$. This follows by integrating $q^\beta(1-q)$. With partial observation, a per-history guarantee requires one admissible action distribution for the observations. At each compatible history it must give full mass to actions that ensure closure there. Let hidden histories and actions range over the unit interval, with closure exactly when their values differ. A uniform draw succeeds almost surely for each fixed history, yet every action has one exceptional history. Measurable selection and time-consistent continuation are required. A posterior mean integrates completion probability over histories.

Exact-differential route

Represent the ledger increment by a one-form on the specified state space. If the form is the differential of a potential Φ, the increment up to each pre-endpoint time equals the change in Φ. Closure at an independently specified endpoint requires the incoming limit of Φ to exist and equal its initial value.

Equation (8.4) uses the terminal-state value when it equals that limit. A nonzero ledger integral around an admissible closed path, or unequal ledger integrals along admissible paths with the same state endpoints, refutes exactness there. An enlarged state space requires independent revalidation.

Equation (8.4). Exact-differential route and loop diagnostic

$$\omega = d\,\Phi \Rightarrow \int_{\gamma_{S\to T}} \omega = \Phi\left(x_T\right) - \Phi\left(x_S\right)$$

$$\oint_{\Gamma} \omega = 0$$

Information retained by a forward realization

If the future depends on the past only through the current state, how much of the accrued balance must that state retain for closure? At one fixed admissible cut, let L be the accrued contribution and R the entire remainder of the same complete stream under Sections 2–3. Let X include all state information used to generate the continuation, including its endpoint process. Suppose L and R are conditionally independent given X and have finite second moments. Put $D = \mathbb{E}\left[(L+R)^2\right]$. Conditional variance decomposition gives

Equation (8.5). Closure error floor under a screening state

$$D = \mathbb{E}\,\mathrm{Var}\left(L \mid X\right) + \mathbb{E}\,\mathrm{Var}\left(R \mid X\right)$$

$$+ \mathbb{E}\left[\left(\mathbb{E}[L \mid X] + \mathbb{E}[R \mid X]\right)^2\right]$$

$$\geq \mathbb{E}\,\mathrm{Var}\left(L \mid X\right).$$

If $R = -L$ almost surely, X must determine L up to null events. With real L and standard-Borel X, the exact conclusion needs no moment assumption: conditional independence and $R = -L$ make each conditional rational-threshold probability equal to its square, forcing the conditional distribution of L to be a point mass. The state can retain this information outside conscious recall. A noisy assay of X can lose information retained by the actual state.

For a Gaussian L with variance $s^2 > 0$, the same screening condition and $0 < D < s^2$ give a lower bound on Shannon mutual information, measured in bits:

Equation (8.6). Gaussian information requirement for closure accuracy

$$I_2(L;X) \geq I_2(L;-R) \geq \frac{1}{2}\log_2 \frac{s^2}{D}.$$

Here $-R$ reconstructs L with squared error D. The first step is data processing for $L \to X \to R$; the second is the Gaussian squared-error bound (Cover & Thomas, 2006). In this range, halving the root-mean-square closure error raises the information lower bound by one bit. A state with at most K values must satisfy $D \geq s^2 / K^2$. The bound is trivial when $D \geq s^2$ and excludes finite K at zero distortion.

9. Completion potential across settings

Completion potential

Let $\mathrm{H}_{i,t}$ be the frozen pre-endpoint filtration completed under P_i^0, with $\overline{\Xi}_{i,t}$ and the current-risk indicator $1_{\{T_i^\star > t\}}$ measurable with respect to $\mathrm{H}_{i,t}$.

For a positive-probability event $A_{i,\varepsilon}$, require on $\{T_i^\star > t\}$ that the following equality hold almost surely:

Equation (9.1). State sufficiency for completion potential

$$P_i^0\left(A_{i,\varepsilon} \mid \mathrm{H}_{i,t}, T_i^\star > t\right) = P_i^0\left(A_{i,\varepsilon} \mid \overline{\Xi}_{i,t}, T_i^\star > t\right) = h_{i,\varepsilon}\left(t, \overline{\Xi}_{i,t}\right).$$

Otherwise augment the state or retain history. Nested bands require potentials monotone in ε. Under a common regular conditional residual law, shrinking the bands gives the exact-atom probability below.

Equation (9.2). Positive-probability completion potential

$$A_{i,\varepsilon} := \{\left|Z_i^\star\right| \le \varepsilon\}$$

$$h_{i,\varepsilon}(t, \xi) := P_i^0\left(A_{i,\varepsilon} \mid \overline{\Xi}_{i,t} = \xi, T_i^\star > t\right)$$

$$\lim_{\varepsilon \downarrow 0} h_{i,\varepsilon}(t, \xi) = P_i^0\left(Z_i^\star = 0 \mid \overline{\Xi}_{i,t} = \xi, T_i^\star > t\right)$$

Let $\mathbb{E}_i^0$ denote expectation under P_i^0. Require $P_i^0\left(A_{i,\varepsilon} \mid \mathrm{H}_{i,0}\right) > 0$ almost surely on the declared risk set. The normalized conditional probability is a nonnegative martingale. Given the history so far, its expected value at a later time equals its current value. Conditioning on a probability-zero event uses a declared regular-conditional bridge, density limit, or other justified

construction (Doob, 1957; Chetrite & Touchette, 2015). Under a reference law supported entirely within the band, the potential is one on almost every at-risk history. A varying potential requires positive probability of missing the band and histories to differ in their completion probabilities.

Within a fixed risk set and time, a history group's completion probability is the overall completion fraction times the group's frequency among completers divided by its frequency among those at risk. The group and the event of completion must each have positive probability in that risk set. Independently sampled records or explicit structural assumptions supply these quantities before testing.

Equation (9.3). Normalized completion-event martingale

$$M_{i,\varepsilon}(t) := \frac{P_i^0(A_{i,\varepsilon} \mid \mathrm{H}_{i,t})}{P_i^0(A_{i,\varepsilon} \mid \mathrm{H}_{i,0})}$$

$$\mathbb{E}_i^0\left[M_{i,\varepsilon}(t) \mid \mathrm{H}_{i,s}\right] = M_{i,\varepsilon}(s), 0 \le s \le t$$

Testing one completion potential across outcomes

Fit a discrete-time Markov reference law and its completion potential on training data. Freeze both fits, the state, normalization, tolerances, support and observation model. When the state captures the history relevant to completion, including endpoint status, weight each transition by the destination's next-time completion probability. When the weighted row sum is positive, divide each entry by that sum. In a coherent model the sum equals the starting state's completion probability. For two

starting states and two destinations at a fixed time, suppose all four cells are positive under both laws. Conditioning preserves their odds ratio (the diagonal product divided by the off-diagonal product) because the row normalizers and destination weights cancel. Score the model and the rival forecast on the same held-out transitions. For a new intervention, apply Section 8’s action support and causal assumptions.

The Doob bridge conditions four independent fair steps of size one to finish at zero. At balance +1 after the first step, the next step returns toward zero with probability 2/3. At the same balance after the third step, return is certain. Under the bridge, the best constant forecast across equally weighted cuts is 5/6. Its expected Brier loss exceeds the bridge forecast’s by 1/36. A horizon-sensitive ordinary rival can reproduce both probabilities. Compare the models on held-out histories selected by cut and balance. Selecting only eventual closers would force the last return under either model. The companion varies the announced horizon at a common first-step cut.

Predictions the mechanism must satisfy together

For a reserve route, test the registered difference in the calibrated proxy’s conditional variance between two regimes, accounting for prognosis. For the registered completion event, compare held-out probabilities with the independently adjudicated binary outcome using the Brier score (Gneiting & Raftery, 2007). Among streams at risk, using the true full-history probability in place of the true state-conditioned

probability reduces expected Brier loss by the mean squared difference between them. Under the state-sufficiency condition in Equation (9.1), that gain is zero. For fitted forecasts, first test whether adding the accrued-ledger summary improves on the frozen baseline. Then add the remaining history and test for score equivalence within the specified margin.

Random-cut complementarity

At randomized cuts, partition the complete stream under Section 4's additive representation. Disjoint half-open intervals assign each cut-time atom once. Obtain remainder observations from post-cut data or a validated terminal-remainder bridge, independently of the tested terminal target.

In Equation (9.4), u_{ik} and v_{ik} are errors in the independently calibrated prefix and remainder channels. Specify a joint observation or bridge law across channels and repeated cuts, allowing within-stream dependence and a separate intercept for each stream. Estimate β freely using prefix variation and Section 12's errors-in-variables conditions. The restriction $\beta = 1$ means a signed prefix coefficient of -1. For the exact direct partition, $\beta = 1$ and $\alpha_i = Z_i^{\star}$ hold for every fixed stream total. For example, totals of zero and plus five have the same slope and different intercepts.

The expected sum of measured prefix and remainder under the within-stream cut law equals the terminal total plus the two mean errors. Subtract their calibrated sum to recover α_i. If only bounds on the combined bias are available, subtract its upper

and lower limits from the expected measured sum to bound the intercept below and above, respectively. An additional latent residual requires a separate bridge or an explicit model of misspecification.

Equation (9.4). Random-cut additivity and stream-specific residual adjudication

$$Z_i^{\star} = L_{ik}^{\mathrm{pre}} + L_{ik}^{\mathrm{post}}$$

$$\hat{L}_{ik}^{\mathrm{pre}} = L_{ik}^{\mathrm{pre}} + u_{ik},$$

$$\hat{L}_{ik}^{\mathrm{post}} = \alpha_i - \beta\, L_{ik}^{\mathrm{pre}} + v_{ik}$$

Joint restrictions across a complete partition

The random-cut relation extends to a fixed finite partition of a complete stream domain. Under a common probability law and representation, let X_j be the signed contribution in part j, assume finite second moments, and set $X = (X_1, \ldots, X_m)^{\top}$, $\mu = \mathbb{E}\, X$, and $\Sigma = \mathrm{Cov}(X)$. Under this exhaustive, nonoverlapping partition, their sum is the terminal residual. Its almost-sure zero-closure condition has the equivalent moment form

Equation (9.5). Complete-partition moment criterion

$$\sum_{j=1}^{m} X_j = 0 \quad \text{almost surely}$$

$$\Leftrightarrow 1^{\top}\mu = 0 \ \text{ and } \ \Sigma 1 = 0.$$

For the forward direction, a zero total has zero mean and zero covariance with every component. Conversely, the two conditions give zero mean and zero variance for the total, so it equals zero almost surely. Each component satisfies

$$\operatorname{Var}(X_j)+\sum_{k\neq j}\operatorname{Cov}(X_j,X_k)=0.$$

The four-step walk above has unconditioned terminal variance four and closes with probability 3/8. Among its six equiprobable closing paths, each step has zero mean and unit variance. Every distinct pair has covariance −1/3, so every covariance row sums to zero. The general row-sum condition permits pairwise covariances of either sign.

Randomized remainder and delayed-treatment localization

Let L denote the signed prefix before assignment. Treatment begins at a common, prespecified time τ after assignment. On the primary branch, E contains post-assignment contributions before both τ and the truth endpoint; F contains the remaining strict-pre-endpoint contributions at or after τ. If the stream ends before or at τ, F = 0. The inclusive branch adds J_T:

$$R = E + F, \qquad R_{\text{incl}} = E + F + J_T.$$

Exact closure on the declared branch makes its complete remainder equal to −L. If assignment is independent of L, remainder distributions agree across arms; stratified randomization requires within-stratum comparisons or standardization to one baseline distribution. A closure-enforcing

simulation must check that its selection or reweighting preserves the assignment-independence premise.

The localization prediction requires a structural model for both named interventions on a common probability space. For the same pre-assignment history h and declared bearer, routing, representation, time cut, and endpoint architecture, suppose primary closure holds under both arms:

Equation (9.6). Delayed-treatment compensation under paired closure

$$L(h)+E^{a}(h)+F^{a}(h)=0,$$

$$L(h) + E^{b}(h) + F^{b}(h) = 0.$$

$$E^{a}(h) - E^{b}(h) = -[F^{a}(h) - F^{b}(h)]$$

An independently justified nonzero difference in the entire later contribution requires an equal and opposite early difference. On the inclusive branch the later contrast includes the arm-specific terminal terms. A directional or distributional biological prediction combines this identity with a history distribution and a validated observation or viability bridge.

For a marker measured between randomization and treatment, a local forward model predicts equal arm distributions when assignment has no earlier causal channel. If that marker also precedes the stream, it tests the developmental extension in Section 14.

For a sharp formal fork, consider one candidate continuation d that ends biological life essentially immediately and another that permits ordinary continuation. Let h now denote the complete predecision history. Under the additional biological-death condition and an independent bridge placing the stream endpoint no later than death, let C_d(h) be the complete decision-to-endpoint contribution for the declared branch. A necessary compatibility condition is:

$$L(h) + C_d(h) = 0.$$

The phenomenally silent case C_d(h) = 0 requires L(h) = 0. Choice probabilities come from a separate selector specifying how admissibility and available information affect action. An independently randomized, locally concealed contingency mapping lies outside an ordinary local policy's information set.

10. Comparing models with the same information

Give each forecaster the same information at prediction time (Blackwell, 1953). Use later observations and endpoint labels for scoring. Match measurements, missingness inputs, training splits, tuning opportunities, complexity and computing budgets. Compare calibration and proper scores on the same held-out cases at prespecified horizons. The reproducibility capsule locates the data and code for the three reanalyses.

Predictions that distinguish the models

A return to baseline can leave a positive accrued balance. Compare a zero-restricted baseline with nonzero attractors specific to a person or state, allowing variability and return rates to differ (Kuppens et al., 2010). Adaptation also varies by life event (Diener et al., 2006; Luhmann et al., 2012). Across 19 cohorts totaling 28,482 participants, mood declined on average during rest and simple tasks (Jangraw et al., 2023). Give every forecaster elapsed time.

In 26 participants, a model of momentary happiness based on rewards, expectations and prediction errors gave a mean forgetting factor of 0.61. At that value a reward retains about 8 percent of its weight five trials later (Rutledge et al., 2014).

Keren et al. (2021) modeled how earlier rewards shape mood. Using 68 public workbooks, we fitted exploratory models on 45 participants and made 782 forecasts of the next recorded 0–1 slider position for the other 23. Adding the earlier within-block mean to the preceding rating and trial index lowered equal-person mean squared error from 0.01019 to 0.00811, with gains

for 19 of 23 people. After inspecting the results, we compared 529 later-block forecasts from all 23. Adjusting both models for prior-block reporting level reduced the gain from 0.001761 to 0.000185 (fixed-fit participant bootstrap, 95% interval −0.000590 to 0.000764). With a space-bar confirmation on the same trial record for each contributing rating, the adjusted gain was 0.000422 across 417 forecasts from 19 people (corresponding interval 0.000146 to 0.000737). The companion documents the unresolved mapping from the 68 files to the article’s 60-person random-task sample.

In CASE (Sharma et al., 2019), we centered film ratings at the joystick midpoint of five. We predicted signed area from 60 to 90 seconds using first-minute area and the mean rating from 55 to 60 seconds. The unrestricted prefix-area slope was +0.115. A nonpositive constraint gave zero, and refitting the intercept and recent-rating slope reproduced the recent-rating-only forecasts exactly. Across 80 participant–film segments from ten held-out people, both had mean squared error 588.74 (rating-point seconds)^2, against 638.46 for the unrestricted model. A post-inspection reconstruction using raw timestamps changed interpolation and window alignment. The tie persisted.

In the MIDUS diaries, current affect, diary day and a stressor indicator predicted next-day negative affect (Ryff & Almeida, 2022). Adding the earlier mean lowered equal-person mean squared error from 0.04273 to 0.03571 across 1,979 verified pairs from 353 people in held-out families. Of 2,048 eligible forecasts for 354 people, 19 lacked verified next-day pairings. Another 50 had verified pairings but invalid next-day scores.

With predictions and equal-person weights fixed, allowing those 69 outcomes any value from 0 to 4 gave attainable full-panel gains from −0.00234 to 0.01728. In an exploratory analysis, unknown scores ranged continuously from zero to a common ceiling. The lower gain bound reached zero at a ceiling of about 2.75934.

For endpoint-conditioned bridges, score event probabilities, conditioning normalizers and the joint hazard–ledger forecast at unexpected endpoints using Section 8's prognostic covariates and Section 12's observation model.

In film experiments, the duration of pleasant or aversive clips had little influence on later evaluations once the real-time ratings were taken into account (Fredrickson & Kahneman, 1993). Compare an accumulated-area forecast with a memory model that weights the peak and the ending (Kahneman et al., 1997). Score predicted area against independently measured area and predicted recall against the later global evaluation.

Low mood predicted activities associated with subsequent mood improvement (Taquet et al., 2016). Give both policies measured activities and test the residual term beyond reward, effort, habit, threat and viability on held-out data.

Let Q be the rival law and $\mathcal{G}_{i,0}^{ex}$ its baseline information sigma-field. Under Q, a standard normal residual independent of the baseline has conditional mean zero and variance one with Q-probability one. This normal residual has zero probability of exact closure. The broader variance condition permits an atom at zero with mass below one.

Equation (10.1). Expected neutrality is weaker than pathwise closure

$$\mathbb{E}_Q\left[\left(Z_i^\star\right)^2\right]<\infty,\ \mathbb{E}_Q\left(Z_i^\star \mid \mathcal{G}_{i,0}^{\mathrm{ex}}\right)=0\ Q\text{-a.s.},$$
$$Q\left(\mathrm{Var}_Q\left(Z_i^\star \mid \mathcal{G}_{i,0}^{\mathrm{ex}}\right)>0\right)>0$$

Identical observable laws under all allowed interventions require a new intervention or measurement to distinguish the mechanisms.

History information versus predictive approximation

Under the held-out law, let B denote baseline information, H additional history, and Y a future outcome. Let p_H and p_0 be the true conditional laws of Y given (B, H) and B, and let q_H and q_0 be frozen densities on the same space, positive on true-law support. Assume finite log expectations. Define

$$\mathcal{E}_H=\mathbb{E}\,D_{\mathrm{KL}}(p_H\,\|\,q_H),\ \mathcal{E}_0=\mathbb{E}\,D_{\mathrm{KL}}(p_0\,\|\,q_0).$$

Equation (10.2). Logarithmic-score gain decomposition

$$\mathbb{E}\log\frac{q_H(Y\mid B,H)}{q_0(Y\mid B)}=I(Y;H\mid B)+\mathcal{E}_0-\mathcal{E}_H.$$

Adding and subtracting the true log densities gives Equation (10.2). A better approximation can give the history model a scoring advantage even when both models target the same conditional law. The expected log-score gain equals conditional mutual information exactly when the approximation errors agree (Cover & Thomas, 2006; Gneiting & Raftery, 2007).

11. Existing evidence and the scientific opportunity

Established neighboring results

In 16 participants at four heat levels, pain intensity and subsequent peak relief had a Spearman correlation of 0.80 across participant-by-condition means (Leknes et al., 2008). Equal peaks can conceal different totals. A calibrated time course could compare accumulated pain and relief.

On the scalar density branch, let a complete stream have finite, positive exposure on both sides of neutral. If exposure above neutral is twice that below, closure requires the exposure-weighted mean valence magnitude below neutral to be twice that above. Across 65,721 reports from 11,572 people, Trampe et al. (2015) estimated 41 percent positive only, 16 percent negative only and 33 percent mixed. The test requires intensity and exposure for each report.

Opponent and allostatic processes model rebound (Solomon & Corbit, 1974; Koob & Le Moal, 2001). Incentive salience distinguishes wanting from liking (Berridge et al., 2009). Interoceptive prediction, homeostatic reinforcement learning and mood as a signal of momentum link past states to behavior in different ways (Barrett & Simmons, 2015; Keramati & Gutkin, 2014; Eldar et al., 2016). Biochemical integral feedback can stabilize a variable by accumulating deviations from its set point (Barkai & Leibler, 1997; Briat et al., 2016; Aoki et al., 2019). Does the corresponding signal in an affective model track the stream’s signed balance?

12. Measurement, missingness, and endpoint validity

Representation before accumulation

At the intended temporal resolution, test whether each person's measurement scale is stable over time. Test invariance separately for comparisons across populations or states (Adolf et al., 2014). Retest the anchors at changes of reporting scale (American Educational Research Association et al., 2014; Krantz et al., 1971; Meredith, 1993). Map shared data and preprocessing, and model joint error before combining channels. Multitrait–multimethod validation and negative controls test whether arousal, salience, motor capacity or demand explains their agreement (Campbell & Fiske, 1959). Triangulation asks whether the same fitted relation predicts results across methods with different expected biases, such as prospective reports and retrospective recall (Lawlor et al., 2016; Munafò & Davey Smith, 2018).

Observation coverage and unsampled intervals

Retain separate marks for scheduled, randomized, participant-triggered, clinician-triggered, safety-triggered, endpoint-triggered and device-failure observations.

Outcome-independent sampling with known positive inclusion probabilities supports design-based estimation of a specified total. Correcting for informative sampling requires positivity and an identified observation process.

On the published CASE grid, we masked the first and last five seconds of each film's opening 90-second window (Sharma et

al., 2019). For participant 1 in film 1, the middle 80 seconds contributed 136.56 rating-point seconds relative to the joystick midpoint of five. The 0.5–9.5 scale limits the missing area to ±45, giving a total between 91.56 and 181.56. Restoring the ratings gives 136.75. Across 240 masked film windows (30 people, eight films each), 128 signed-area intervals excluded zero at that midpoint and 90 did so for every midpoint within 0.25 rating points of five.

When unobserved experience affects reporting

Unobserved valence and proximity to the endpoint can influence reporting through fatigue, delirium, sedation, communication failure, withdrawal, care transitions or device loss. Prochnow et al. (2025) modeled whether each prompt was completed in a 12-month study. The mean participant completion rate was 77 percent in the 207-person analysis, and 123 of the 246 fully enrolled people finished the year. Model missed observations and study exit separately. Report how estimates, bounds or decisions change across prespecified selection and pattern-mixture models (Rubin, 1976; Scharfstein et al., 1999; Little & Rubin, 2019).

Measurement burden and reactivity

Measurement burden and emotional reactivity enter the trajectory as intercurrent events. In a randomized 14-day student study, 60-item questionnaires produced more burden, lower compliance and more self-reported inattentive responding than

30-item questionnaires. Increasing prompts from three to nine daily did not produce the same pattern (Eisele et al., 2022). The proportion of within-person item means equal to zero was 28% lower with visual analogue ratings than with seven-point Likert ratings (Haslbeck et al., 2025). Fixed prompts yielded lower affect variability than random prompts in eight of 16 models (Langford et al., 2026). Calibrate questionnaire length, prompt schedule, response format, anchors and interface defaults against response quality, missingness and burden in the target population.

Replicate channels and recoverability

Suppose two readings measure one latent quantity with additive error. If the quantity and both errors are mutually independent and the readings' joint characteristic function is nowhere zero, their joint law identifies all three distributions up to a location shift (Kotlarski, 1967). Fixing one component's location by independent calibration resolves that shift. For the Day Reconstruction Method, model which episodes people recall from the preceding day and how they rate those episodes (Stone et al., 2006). Among 105 rheumatology patients, paired momentary and end-of-day reports showed strong daily correspondence for pain and item-specific level offsets (Broderick et al., 2009). For the signed duration-weighted functional, compare channels on the same windows within each person. Estimate the between-method mean difference and limits of agreement for repeated pairs, allowing for within-person

dependence and changes in the underlying state (Bland & Altman, 2007). For an additive error, observed terminal variance equals latent variance plus error variance plus twice their covariance.

Suppose two channels measure the same complete-stream residual plus error, with finite second moments. If both errors and their product have conditional mean zero given that residual, the expected product of the uncentered readings equals the residual's second moment. A positive lower confidence bound for this moment challenges probability-one closure in the sampled population. For bounded shared error, see the companion's design benchmark, Section 10.

Shared measurement error can itself mimic correction. In discrete time, suppose zero-mean errors e_k are mutually independent, independent of the latent process, and have common variance σ^2. For fixed weights w_k and a positive integer lag h,

$$\operatorname{Cov}\left(\sum_{k \le t} w_k e_k, e_{t+h} - e_t\right) = -w_t \sigma^2.$$

Only current error is shared, with its prefix weight and reversed sign. For dependent or unequal-variance errors, use the corresponding covariance terms. For Equation (9.4), an instrument correlated with the true prefix within a stream and uncorrelated with both channel errors can identify the slope. Calibration data or explicit measurement assumptions can identify or bound the joint error moments.

13. Confirmatory decision structure and model recovery

Confirmatory reporting and decision vocabulary

The decision record specifies the bearer and complete domain (Sections 2–3), representation and exposure (Sections 2 and 4), the endpoint (Section 7) and the observation model (Section 12).

Record the target geometry (exact residual, finite band or atom at zero) and modality ([SET-ALL], [P1] for probability-one, [POLICY-ALL], [SUPPORT], [CONTINUATION-ALL] or [FINITE-CLASS]). For [SUPPORT], record the specification from Section 3. For [FINITE-CLASS], name the class, its quantifier and the members sampled or the enumeration or argument covering the class.

For TargetStatus use Equivalent, Different / off-band, Informative but inconclusive, No adequate estimate / invalid gate, Target-adverse, or Not adjudicated. For valid, nonempty truth-containing sets, use Target-adverse for exact-zero exclusion and Different / off-band for disjointness from the registered finite band. For RouteStatus and ComparatorStatus use Route-favored, Rival-favored / Route-failed, Model-ambiguous, Open-set inadequate, No adequate estimate / invalid gate, or Not assessed. Not assessed marks a comparison outside the registered scope.

RegimeDomain identifies the intervention or intervention set. RegimeQuantifier distinguishes one named intervention, all members of a named finite set, every intervention in a declared structural domain, and outside scope. RoutePolicyAttainment distinguishes fixed-policy probability-one completion, almost-

sure attainability, limit-sure only, positive-probability only, and unavailable attainment.

For the evidence object, record ObjectType (truth-containing outer set, confidence set, posterior or sensitivity region), ObjectStatus (valid, empty/incompatible, unresolved, unavailable or uncomputed), ObjectReason, PrerequisiteStatus (valid, failed or unavailable), and Guarantee. InferenceStatus specifies identified-set, frequentist, Bayesian or sensitivity-based inference. A failed or unavailable prerequisite gives No adequate estimate / invalid gate for the affected verdict. A certified empty feasible set gives ObjectStatus = empty/incompatible and TargetStatus = Not adjudicated.

Recovery before interpretation

The blinded recovery suite combines the routes and rivals in Sections 8–10 with the observation and residual-law families in Sections 5, 12 and 16. Include synthetic and semi-synthetic mixtures, flexible baselines, heavy tails, stochastic volatility and open-set alternatives. In the companion's constructed benchmark, two forecast gains and a signed effect select the generating model with probability 88.65% under either specified law for 400 independent people observed once each.

Simulate each generator through the study's allocation, observation, missingness and selection pipeline across independently justified parameter ranges. Let signed experiences be successive state changes. With independent standard normal

states, the cut state enters the prefix and immediate remainder with opposite signs. At fixed current experience, a larger prefix predicts a smaller remainder despite an unrestricted terminal total. For a persistent rival, start at zero and let each state equal 0.8 times the last plus independent standard normal noise. Given the cut balance, the second and third subsequent increments sum in expectation to −0.288 times that balance. The total has a continuous distribution at every fixed positive horizon. Report parameter recovery, a generating-model by fitted-model confusion matrix, and a second matrix comparing known and reported scientific verdicts (Wilson & Collins, 2019). Prespecify an acceptable false-closure rate and required power for a meaningful nonzero residual. Report simulation replication counts, Monte Carlo uncertainty for performance estimates, worst-class performance and held-out predictive adequacy (Morris et al., 2019).

Negative controls and open-set alternatives

Assess absolute fit with negative controls for construction artifacts and shared causes (Section 12; Lipsitch et al., 2010). Record Open-set inadequate if every offered model fails and Model-ambiguous if adequate models remain inseparable.

Register each new rival’s equations and retirement rules before testing it. If the analysis repeatedly misses a mechanism in data generated under its own equations, revise the assay before using it to judge that mechanism. Retire the mechanism on the tested domain after a replicated, adequately powered

failure of its defining prediction or a prespecified decisive loss to an equally informed rival.

Bounds for a partially classified frame

Retain all eligible unresolved bearer, endpoint, observation and bridge cases under one target, modality, eligible frame and nonnegative weighting rule. Truth-containing favorable and adverse classifications under common gates make Equation (13.1) a deterministic bound. Given only these classifications and weights, assigning all unresolved cases jointly to the favorable or adverse category attains the lower or upper endpoint, respectively. Further joint restrictions can narrow the range. Confidence-set versions require simultaneous coverage. Label purely descriptive ranges.

Equation (13.1). Transparent unresolved-case bound

$$W_E := W_F + W_A + W_U > 0,$$
$$\frac{W_A}{W_E} \le p_{\text{viol}} \le \frac{W_A + W_U}{W_E}$$

Let W_F be weighted favorable classified mass, W_A weighted adverse classified mass, W_U weighted unresolved or gate-invalid mass, W_E total eligible mass, and p_{viol} the violation fraction in the defined weighted eligible frame. Beyond this frame, design-based inference requires stream inclusion probabilities or design weights and must propagate sampling uncertainty under that design (Horvitz & Thompson, 1952). For model-based generalization, specify the target population and the assumptions linking it to the observed streams (Valliant, 2020).

14. Experimental program: calibration and mosquito terminal-choice tests

Stage 0 — protocol fixation, public preregistration, and rigorous design

Fix the confirmatory definitions, models and contrasts before outcome access. Public preregistration adds an external timestamp and persistent identifier. Release simulation labels and empirical outcomes to the masked analysts and adjudicators after their respective pipelines are locked.

Stage 1 — representation and local complementarity

Randomize benign assignments within bounded episodes across valence modalities, controllability, predictability, duration and horizon (Klasnja et al., 2015). Under consistency, well-defined interventions, positivity and the declared interference structure, randomization identifies assignment effects. Estimating delivered, experienced or latent phenomenal dose requires the calibrated measurement and noncompliance models.

Calibrate each participant's neutral anchor, exchange between signs and positive reference area in a separate session. Observe bounded episodes through two channels. At a randomized cut, predict signed area from one increment after the cut to the scheduled end. The intervening increment counts toward the full episode ledger. Score all three forecasts against the independently calibrated outcome channel over the delayed window. Determine eligibility at the cut and retain missing windows and their durations under Section 12's observation model. Hold out whole participants and, when generalizing to

new stimuli, whole stimulus families. Account for both sources of variation (Judd et al., 2012).

All three forecasters receive the pre-cut history, calibrated prefix, recent state, activities, elapsed time and known remaining duration. Include the earlier-block reporting level with a training-only fallback. At fixed state and horizon, the change in mean future area predicted by compensation is minus the fitted fraction times the change in calibrated prefix area. Prespecify a closed nonnegative range for the fraction that includes zero and permits only feasible shifts at that horizon. Refit nuisance coefficients at zero for the second forecast. Fit a named ordinary process, such as the state-difference model in Section 13, to the same training episodes. Use the same delayed target for training comparisons, planted-data recovery and held-out scoring. Under Section 13's independent-state generator, the delayed area has conditional mean zero. Register the direction and minimum size of the shift that compensation predicts at the same cut (Myung & Pitt, 2009). Freeze all three fits and the clock-only reference before scoring. Report the reference separately from the two primary comparisons.

Within each person, average the rival's squared error minus the candidate's across eligible cuts. Divide by the square of the person's reference area, then average equally across people. The resulting dimensionless gain is invariant to a positive change of within-person units. Rescale the positive eligibility floor with the reference and record exclusions. Compare raw errors as a secondary outcome only in a common calibrated unit.

Use pilot repeatability and a prespecified worthwhile margin for each comparison to size the study. Report the observed-channel gain. For latent area, vary channel offsets, shared bias, missing exposure and reference uncertainty jointly across every outcome, forecast and weight they affect. For fixed forecasts, the difference in squared errors is affine in the true signed area of the scoring window. Before recruitment, use planted training data to fit the forecasts. With predictions and weights frozen, maximize the smaller excess above margin over one common set of admissible outcomes. A positive maximum means that both point-score margins can be exceeded together.

Condition inference on the frozen fits and training split. Construct simultaneous confidence limits covering both population gain ranges, including calibration uncertainty. The companion gives conservative limits for independent participants with bounded scores. Check coverage at tied optima (Fang & Santos, 2019). Each comparison favors the candidate if its interval lies strictly above the positive margin and the rival if it lies strictly below the negative margin. An interval contained in the closed equivalence band supports practical equivalence. All other comparisons remain unresolved.

Apply Section 13’s blinded recovery procedure to the joint decision, including small effects, rounding and observation loss. Record each rival’s simultaneous gain interval, margin and interval decision. For two adequate models, ComparatorStatus is Route-favored for a candidate win, Rival-favored / Route-failed for a rival win, and Model-ambiguous for equivalence or an unresolved comparison. Assign RouteStatus in favor of

compensation when an active effect transfers with the registered sign and magnitude and both gains clear their margins under the joint limits. Assign the registered TargetStatus to the bounded episode. A whole-stream assessment also measures or bounds the contributions outside that window (Section 4).

Stage 2 — frozen completion-geometry transport

Test Section 9's frozen predictions at an independent site or through an independent measurement channel. Report which changed. Hold out randomized-information actions, conditional increments and independently scored future area. For off-support actions give identified bounds or No adequate estimate.

Immediate analyses in existing data

For endpoint prediction in an at-risk cohort, fit representations, covariates, competing-risk forecasts and tuning out of fold at each landmark. Score observed endpoint time and cause before route-by-surprise comparisons. Use held-out compensator-residual and time-rescaling diagnostics matched to the ordinary, marked or cause-specific, terminating and right-censored process (Cox, 1972; Andersen et al., 1993; Brown et al., 2002). Distinguish cause-specific intensity, cumulative incidence or event probability, and interventional endpoint risk.

Recheck Section 12's measurement calibration in each held-out setting and specify which causal relations must remain stable across populations (Pearl & Bareinboim, 2014).

Randomized stopping-kernel experiment

Nonanticipating stopping kernels randomized from frozen histories probe occupation functionals on a prespecified basis. Apply Stage 1's intervention and interference conditions. Use known assignment probabilities or justify conditional exchangeability of future potential outcomes, with common support and explicit observation and censoring models. Report the basis, rank, condition number and null space. When the kernels separate the basis, estimate its coefficients. Otherwise estimate stable functionals or valid bounds. Distinguish physical stopping from a disclosed horizon.

Mosquito terminal-choice and future-regime experiment

The mosquito program tests whether prior conditions and time since a salient experience predict continuing-life choices, and whether future adult-history assignments predict earlier developmental outcomes.

Species, developmental stages, and realistic timing

Use a specified noninfected laboratory strain of *Aedes aegypti* in a contained insectary. Eggs hatch into aquatic larvae, not flying adults; larvae pass through the pupal stage before adult emergence. The CDC describes approximately seven to ten days from egg to adult under favorable conditions, while actual timing depends on the colony and its conditions (Centers for Disease Control and Prevention, 2024). Distinguish egg hatching, adult emergence, chronological age, adult age, and

treatment duration in the records. Egg-storage age is not a substitute for developmental age.

Following adult emergence, maintain a common seventy-two-hour maturation period before any assigned adult-history exposure. This design choice is informed by evidence that flight performance matures over the first one to three adult days (Briegel et al., 2001). Verify ordinary flight and handling performance in independent calibration cohorts. Published laboratory methods provide the species-specific husbandry and containment foundation (Wohl & McMeniman, 2023). Prespecify sex and mating status, use balanced allocation across emergence batches, and analyze sex-specific responses rather than assuming identical physiology. The study uses sugar-resource manipulations.

Female *Aedes aegypti* retain learned odor–blood-reward associations for at least 24 hours (Vinauger et al., 2014), while sugar pairing can make DEET odor an appetitive cue (Lazzari et al., 2026). Aversive odor learning also persists for 24 hours and depends on dopamine-1 receptor signaling (Vinauger et al., 2018). These findings provide tractable assays for calibrating learned preference and its persistence in independent cohorts.

A biologically realistic duration comparison tests adults at six days after emergence. Assign adult-history durations of 24, 48, or 72 hours. For the 24-hour condition, begin the history on adult day five; for 48 hours, on day four; for 72 hours, on day three. Keep the preceding period under the same common husbandry. All groups then reach the chamber at the same adult age and prespecified circadian phase. Retain the actual elapsed

times and pre-treatment histories, since changing onset time changes an exposure history, not an abstract dose alone.

The initial 300-egg feasibility cohort should evaluate one prespecified duration and post-event schedule, for example the 48-hour adult-history condition. Further independently randomized cohorts evaluate the other durations and post-event schedules. With seven to ten days of development, testing at adult day six falls about thirteen to sixteen days after development begins. Observed emergence and completion of maturation determine eligibility.

Allocation before hatching and the three histories

Record the 100/100/100 future assignments before hatching using a concealed randomized allocation. Block across parental or egg-laying batches where feasible. All eggs, larvae, pupae, and newly emerged adults receive equivalent physical conditions until their assigned adult treatment begins. Balance containers, density, water, food, handling, position, temperature, humidity, and light exposure, and record departures. Use coded assignments so staff maintaining the common developmental environment need not know future history labels. Use multiple independent containers per history.

Group 1 receives normal laboratory living conditions. Group 2 receives enriched conditions selected to provide favorable temperature, reliable food, and other species-relevant opportunities. Group 3 receives an adverse-history condition involving less favorable, prospectively specified resource or

environmental circumstances. The operational differences and exposure limits are fixed through species-specific calibration and review before confirmatory outcomes are examined. Measure flight capacity and pre-choice attrition separately. Calibrate each sex with graded sucrose-choice and feeding-acceptance assays: *Aedes aegypti* acceptance varies with sugar concentration and composition, while some paired-diet responses differ by sex (Ignell et al., 2010). Normal, enriched, and adverse designate assigned environments; their intended phenomenal signs require independent calibration.

Retain counts for eggs allocated, eggs hatched, larvae reaching pupation, adults emerging, adults reaching the scheduled test, and adults producing an observed choice. Record pre-choice mortality, nonchoice, handling failures, and observation losses without substituting animals after outcomes become known. Container-level treatment makes the container an experimental replicate; mosquitoes sharing it are nested observations (Lazic et al., 2018).

The twenty-outlet chamber and the five-percent null

Test one mature adult at a time in an enclosed chamber with twenty equivalent entrance apertures. One outlet leads to normal continuing-life conditions; the other nineteen lead to a contained terminal apparatus. The proposed implementation retains the mosquito-zapper concept, subject to independent validation of the actual terminal procedure, its latency, and its suitability for the proposed study. The relevant decision is the first

prospectively defined committed outlet entry before feedback or contact with the downstream apparatus. Record route entry, subsequent survival, verified terminal events, and their times as separate outcomes.

Choose the continuing-life outlet uniformly and independently for each trial and conceal its physical identity. A separately chosen blocked allocation must instead record and analyze its actual conditional assignment probabilities. Match geometry, illumination, airflow, humidity, odors, sound, heat, vibration, and any apparatus-associated electrical or chemical cues at the choice point. Use nonterminal trials to quantify side preferences. Independently validate cue concealment with the terminal equipment active. Score trials blind against a frozen route-to-contingency record. The construction must prevent downstream events from broadcasting informative cues to later trial animals.

Let $D=1$ indicate a recorded committed choice before the fixed deadline, let C identify that outlet or its matched holding compartment, and let K be the randomly assigned continuing-life outlet. In the pre-choice arm the null requires K to be uniform within pretrial strata, independently of whether a choice is recorded and, among recorded choices, which outlet is entered. For a nonzero choice probability, the null therefore gives

$$\Pr(C=K\mid D=1)=\frac{1}{20},\Pr(D=1,C=K)=\frac{\Pr(D=1)}{20}.$$

For both timing arms, repeat matched, masked all-nonterminal sham and randomized-label controls; in each sham trial, every

outlet leads to ordinary continuing conditions and one is independently labeled as the analysis target. The hypothesis of interest is a history-related enrichment that depends on the real downstream contingency, not merely on altered locomotion, general escape behavior, or increased willingness to choose an outlet.

Temporally decoupled post-commitment arm. Each entrance leads to a matched passive holding compartment. A video-verified first committed entry and compartment closure precede a freshly generated uniform label K identifying the single continuing-life assignment. The control process logs the commitment timestamp, closure timestamp, randomization input, generated K and subsequent outcome. The randomness, recording logic, segregation of operators and outcome handling must be independently audited.

In the pre-choice arm, K exists before commitment and a route-selection mechanism is evaluated while cue and mapping leakage remain empirical rivals. A PTVC prediction of an association between choice and label in the post-commitment arm requires a separately declared whole-trajectory, nonlocal, or global-selection extension. An association that survives independent randomization, timing and data-integrity audits would challenge the stipulated joint-independence model.

Sharp random-label null. Let H denote the pretrial stratum and c a recorded choice with positive probability in it. The pre-choice allocation and recording assumptions give the first conditional law below. In the post-commitment arm, draw the fresh label uniformly given the locked choice and the full history

at the draw. Averaging over that history within H and c gives the same law. Sum over supported choices for the recorded-choice result. The all-tested match indicator is one for a recorded match and zero otherwise, including a nonchoice for which no label is drawn. Averaging its conditional mean over H gives the marginal all-tested rate.

$$\Pr(K = k \mid C = c, D = 1, H) = \frac{1}{N}$$

$$\Pr(C = K \mid D = 1, H) = \frac{1}{N}$$

$$\Pr(Y_{\text{match}} = 1 \mid H) = \frac{\Pr(D = 1 \mid H)}{N}$$

Constructed joint-law benchmark. Among recorded choosers within each pretrial stratum H, a joint-law family varies choice–label dependence while preserving the specified choice distribution and a uniform later-label distribution. Matching ranges from the independent reference to its sharp ceiling: one minus the total-variation distance between the choice distribution and uniformity. Thus concentrated choices cap matching even when the label histogram remains uniform. The computational companion gives the construction and proof.

Outlet-number transport. Extensions of the twenty-outlet assay to 100 and 1,000 outlets have reference matching rates of 1% and 0.1%, respectively, under the independent uniform-label null among recorded choosers. At each count, establish that adults can reach and discriminate among the entrances and make a valid commitment. Calibrate search time, nonchoice, route availability and accessible cues. Verify concealment,

containment and feasible sample size before testing whether the prediction transfers.

Confirmatory design record. Within each timing arm and history group, randomize comparable units to the terminal or all-nonterminal sham condition. Record the allocation probabilities at the unit randomized, using the block for whole-block assignments. Register the housing-treatment unit, measured animal, eligibility frame, time origin, commitment rule, assignment time, likelihood, dependence model and validity criteria. Each prediction specifies its calibration source, parameter values or bounds, predicted rate or interval, null comparator and decision rule. Record the post-commitment arm's alternative joint choice–label law separately.

The primary outcome is the match rate among recorded choosers. Within each timing arm, compare enriched history with normal history and adverse history with normal history separately. From each terminal-contingency difference subtract its all-nonterminal sham counterpart, using the same outcome definition, eligibility frame and weights. Preregister the normal-history reference, the direction and minimum enrichment for each history, duration and feeding-delay interactions, the early-versus-late feeding contrast, and the direction of one primary developmental contrast. Designate any additional developmental comparisons and allocate the family error budget across both primary history comparisons, timing arms and other confirmatory tests.

What the Law-specific prediction requires

Suppose the animal carries a qualifying stream and the declared biological-death condition applies. An assigned history can leave some streams with a residual that none of the terminal routes can offset in time. If action selection can preserve closure, either the enriched or adverse group can show higher continuing-life choice rates when it contains more such streams than the normal group, with the other choice probabilities held fixed.

Among animals with a recorded choice, let u be the proportion for which the terminal routes are incompatible under the declared model, let s be their probability of selecting the continuing-life route, and let q_0 be the calibrated probability among the remaining animals. If this subgroup is present, every recorded choice delivers its assigned outcome and every stream closes, all members must select the continuing-life route. A selector probability below one describes a sensitivity model with imperfect classification, imperfect delivery or departures from closure. With q_0 = 0.05, rates of 15–25% correspond to u of about 10.5–21.1% when s = 1. If every recorded chooser belongs to the incompatible subgroup and delivery is faithful, exact closure predicts a 100% continuing-life choice rate.

$$p_{\text{safe}} = (1-u)\,q_0 + u\,s\,;$$
$$s = 1\,,\, q_0 = 0.05 \Rightarrow p_{\text{safe}} = 0.05 + 0.95\,u$$

Timing after feeding or another salient event

Adult-history duration and time after a salient event are distinct variables. In an additional series, hold adult age, history duration, and chamber circadian phase fixed while randomizing standardized sugar-feeding opportunity schedules. A proposed starting set places the end of that opportunity 15 minutes, 2 hours, 6 hours, or 24 hours before chamber entry. Record actual ingestion, its completion time, refusal, consumed amount, and subsequent resource access. Validate intake measurement in independent calibration cohorts: a modified flyPAD resolves feeding microstructure in female *Aedes aegypti*, and fluorescent-tracer assays estimate ingested volume. Sucrose intake correlated with recorded sip and activity-bout counts (Henriques-Santos et al., 2023). The primary comparison follows the assigned schedule. Analyses indexed by actual ingestion time or dose must address selection into feeding. An event-triggered alternative can randomize a delay after observed ingestion and specifies its own age and circadian design.

The timing conjecture is that a recent positive experience could temporarily increase an unresolved positive contribution, leaving less time for an opposite contribution before the terminal fork. Under an independently calibrated transient-compensation model, an earlier chamber encounter would therefore produce greater safe-route enrichment than a later encounter. In the hypothetical example of a peak at 1 hour 32 minutes and a test at 1 hour 35 minutes, the informative quantity is the three-minute interval from an independently identified event.

On the positive-residual branch, let t_0 be the event cut and let Δ be an independently supported upper bound on time to the stream endpoint on a terminal route, including waiting, chamber choice and apparatus-to-endpoint latency. For each admissible endpoint u and compatible history h in this window, suppose every admissible continuation at that endpoint and history has absolutely continuous opposite-sign contribution with the same independently calibrated rate bound M, so its total capacity is bounded by the rate times elapsed time and hence by $M\Delta$. If a branch admits additional singular-continuous or atomic negative mass, include an independent bound on that mass in capacity. Count any permitted endpoint atom once. The residual must exceed the combined bound for the same obstruction to follow.

$$\mathrm{Cap}^{-}(t_0, u \mid h) \le M(u - t_0) \le M\Delta, t_0 \le u \le t_0 + \Delta$$
$$L(t_0) > M\Delta \Rightarrow L(t_0) - \mathrm{Cap}^{-}(t_0, u \mid h) > 0$$

A positive event might increase an existing positive residual or reduce an earlier negative one; the proposed direction therefore follows the declared path model.

Feeding also changes ordinary physiology and behavior. Blood feeding can change host attraction, while olfactory sensitivity and activity vary over the daily cycle (Dou et al., 2024; Eilerts et al., 2018). Thus satiation, flight energy, meal mass, motivation, digestion, and circadian phase are explicit alternative explanations. Rotate schedules across independent blocks and measure locomotor competence in separate control cohorts. Varying a feeding delay can change the phase of feeding even when the phase of choice is fixed; record both.

Hatching and adult-emergence predictions

The primary early endpoint can be the proportion of the original eggs hatching within a fixed observation window. Record time to hatching, later adult emergence, and developmental attrition separately. The window and treatment-onset schedule must ensure that the future adult history has not begun when the early endpoint is measured.

Under an ordinary forward-causal null with valid randomization and no physical pathway from the future assignment to incubation, hatching probabilities should not systematically depend on that assignment. The proposed developmental extension asks whether a whole-trajectory or viability constraint changes which individuals enter particular future circumstances. For example, it may predict fewer successful hatchings in the enriched or adverse future groups when those trajectories would otherwise encounter the terminal fork with unresolved contributions.

A credible early effect must survive checks for parental and batch imbalance, container microclimate, label leakage, handler behavior, egg age, fertilization, and differential observation. Use randomization-based comparisons respecting the actual allocation blocks, and repeat with new independently handled batches and analysts. Use post-outcome placebo labels to diagnose analysis errors and independent physical and allocation audits to test channels present during the experiment.

Use all originally assigned eggs as the developmental denominator, including those that never reach the adult

experiment. A model indexed by the future regime actually delivered must separately address cancelled schedules, crossover, non-delivery, and failure to reach adulthood (Hernán & Robins, 2020). Conditioning only on successful emergence can select different animals in different groups. Include safe-route choices in the complete assigned-egg outcome sequence.

Analysis, repetition, and decisive interpretation

Estimate registered history contrasts in designated-outlet matching among recorded choosers in each timing arm. Among all tested adults, report the proportions with a recorded match, a recorded nonmatch or no recorded choice. Distinguish confirmed nonchoice from observation loss in the last group. Conditioning on a recorded choice can change the history groups being compared. Report absolute rates, risk differences, interval estimates, choice latency, and verified post-choice outcomes. Section 16 supplies a sequential option for repeated cohorts.

For 100 independent recorded choices in a twenty-outlet test, the exact 5% null expects five matches. A one-sample, one-sided exact binomial test at a nominal 5% level rejects that null at ten or more matches (actual null-tail probability about 0.0282); its power is approximately 94.5% if the true match probability is 15%. Set the final sample size from the true assignment unit, recorded-choice rate, overdispersion, intended contrasts and multiplicity. Randomization, masking, exclusions and experimental-unit reporting follow ARRIVE (Percie du Sert

et al., 2020). Model the linked assignment sequence as correlated stages.

Choice-model likelihood comparison. For n independent recorded commitments, compare the match probability under independent uniform label allocation, p_0=1/N, with a prospectively specified point alternative p_1. For k matches, the displayed base-ten log likelihood ratio is also the base-ten log Bayes factor favoring the specified alternative over the null.

$$\log_{10} \Lambda = k \log_{10} \frac{p_1}{p_0} + (n-k) \log_{10} \frac{1-p_1}{1-p_0},$$

$$p_0 = \frac{1}{N}$$

Synthetic illustration. Let n = 20,000 independent recorded choices, N = 100, k = 5,000 matches, p_0 = 0.01 and the prespecified point alternative p_1 = 0.25. The base-ten log likelihood ratio is approximately 5,181.09. For k = 19,800 with the same p_1, it is approximately 27,655.10. Relative preference and absolute fit answer different questions: the 25 percent model fits the 99 percent match rate poorly despite its large likelihood ratio against the one-percent null.

Under the one-thousand-outlet transport requirements above, suppose normal-history animals remain near the 0.1 percent reference rate while adverse-history or enriched-history animals select the continuing-life exit 15–50 percent of the time. The enrichment would be 150–500 fold. With uniform labels within each pretrial stratum among recorded choosers, the two ends of the match-rate range require at least 0.886 and 3.983 bits of choice–label information, respectively. This applies the event

bound in Equation (6.3) to matching with base-two logarithms. If cues fully mediate the pre-choice association conditional on the same strata and recorded-choice selection, they must collectively carry at least that much information about the label. A Law of Fairness mechanism that predicts the size and direction of the held-out history and timing contrasts better than its rivals in independent replications would gain empirical support as a route to terminal balance.

15. Endpoint-aligned extension and complete-stream adjudication

Endpoint assessment in longitudinal studies

Across five longitudinal studies, 1,491 of the 3,774 deceased participants with at least three depressive-symptom assessments had one within three years of death. In this group, symptoms rose more rapidly on average beginning about four years before death (Petkus et al., 2025). Among 10,752 cancer decedents with symptom assessments, mean pain and depression scores stayed relatively stable in the final six months while breathlessness and drowsiness worsened (Seow et al., 2011). In the Rush Memory and Aging Project, eventual decedents had lower mean purpose-in-life scores than survivors as early as 13 years before death (Wang et al., 2025). An independent committee adjudicates the end of qualifying experience under Section 7's rules. Compare endpoint revision, residual variance and bounds under joint endpoint, observation and missingness models against symptom, function, prognosis and channel-shutdown forecasts.

Vogel et al. (2013) followed positive and negative affect across five waves over 15 years in 1,671 decedents in the Longitudinal Aging Study Amsterdam. These records describe trajectories retrospectively aligned to death. A prospective forecast starts with everyone at risk at the landmark, even those whose eventual date of death is unknown (Kurland et al., 2009). An ordinary rival fits health trajectories backward from death on the training cohort. At each held-out landmark, integrate over unknown remaining lifetime and compare predictions for the

same observable outcome using only the history then available (Dempsey & McCullagh, 2018).

Clinical estimands

To bound the complete stream's residual at its latent endpoint, project the joint calibration and exposure constraints over the committee's admissible endpoint times or use a validated terminal-remainder bridge. Carry measurement and numerical uncertainty through the calculation. A nonempty feasible set can exclude zero even when its enclosing interval contains zero. One jointly admissible combination of values yielding zero establishes compatibility. Exclusion requires a certificate covering the entire set. Until the search establishes either result, compatibility remains unresolved. After certified exclusion from a nonempty set, report the infimum correction in each prespecified direction and whether it is attained. Mark a direction infeasible if none of its admissible corrections restores closure. An empty feasible set receives Section 13's incompatible-model classification.

Secondary estimands are atom location and mass under Section 16's rule, endpoint dependence and occupation (Sections 6–7), local signed-dose calibration and transport of the completion forecast to the clinical endpoint.

16. Exactness, atoms, and symmetric error control

Equation (16.1). Atom at zero versus no atom at zero statistical branch

$$P_i^Z(dz) = \pi_{0,i}\,\delta_0(dz) + (1-\pi_{0,i})\,G_i(dz),$$
$$G_i(\{0\}) = 0,\ 0 \le \pi_{0,i} \le 1$$

The remainder G_i may contain off-zero atoms, lattice or quantized structure, singular-continuous or absolutely continuous mass, and mixtures. The atom and probability-one branches correspond to $\pi_{0,i}>0$ and $\pi_{0,i}=1$, respectively. For the location-free rival, select a largest atom in the prespecified residual range. Its position and mass define a one-atom representation whose remainder assigns zero mass to that point. Report all tied maximizing locations, or an empty selection if the range contains no atom. Score atom existence and zero location separately. Estimating a stream’s atom mass from its single realized endpoint requires the ensemble or structural model specified in Section 5.

When additive error is independent of the residual, compare characteristic functions of the readings and matched error-only observations at two nonzero frequencies with an irrational ratio. If the error transform is nonzero at both, equality of the population transforms at either frequency confines the residual to integer multiples of the corresponding period. The two sets of multiples intersect only at zero, so equality at both characterizes probability-one closure. Section 10 of the companion’s design benchmark gives the two-frequency test and its finite-resolution controls.

Boundary-aware inference

Both $\pi_{0,i}=0$ and $\pi_{0,i}=1$ are nonregular boundaries, so ordinary interior asymptotics may fail (Self & Liang, 1987); at $\pi_{0,i}=1$ the remainder law G_i is unidentified. Use a procedure justified for the tested boundary and nuisance-parameter class (Andrews, 2000; Davies, 1977). For a bootstrap test, simulate under the specified null and repeat the full fit, including any atom-location search, in each replicate. Measure size across registered nulls and power against alternatives covering zero and location-free atoms, near-zero diffuse laws, off-zero discrete and singular laws, and mixtures near unit mass. Vary rounding, smoothing and detection floors within Section 12's channel model.

Nested-resolution discipline

Prespecify the nested acceptance sets and their limiting rule for one latent target, bridge family and recovery guarantee. When resolution, instrument or model changes, revalidate the target at the new precision using independent channels and calibrated transport.

Symmetric error-control budget

For statistical declarations, the confirmatory multiplicity budget controls false favorable support across representation, origin, routing, endpoint, missingness, route, rival, tolerance and subgroup searches. It also controls false counterexample declarations across streams, endpoints, instruments and models.

Sequential evidence across repeated cohorts

Let E_n be nonnegative and adapted to the recorded data filtration, with $E_0 = 1$, and a supermartingale under every law in the declared null family $\mathcal{H}_0$. For $0 < \alpha < 1$, Ville's inequality gives

Equation (16.2). Time-uniform evidence threshold

$$\sup_{Q \in \mathcal{H}_0} Q\left(\sup_{n \geq 0} E_n \geq \frac{1}{\alpha}\right) \leq \alpha .$$

For any positive level below the stated boundary, stop at its first crossing or the finite horizon, whichever comes first. Nonnegativity and the stopped expectation bound put the crossing probability at or below the reciprocal of that level. First increase the horizon, then let the levels rise to the boundary to obtain Equation (16.2). Confidence sequences provide the corresponding estimation approach (Howard et al., 2021).

For a concrete one-sided choice test, let Y_n indicate a continuing-life outlet choice among the prospectively defined recorded choosers.

If the conditional null gives $\mathbb{E}(Y_n \mid \mathcal{F}_{n-1}) \leq q_n$, with known predictable $0 < q_n < 1$, choose a predictable $0 \leq \lambda_n \leq 1 / q_n$ and use

Equation (16.3). Predictable update for one-sided sequential evidence

$$E_n = E_{n-1}\{1 + \lambda_n (Y_n - q_n)\}.$$

The factor in Equation (16.3) is nonnegative and has conditional expectation at most one. For eligible recorded choices under Section 14’s independent uniform twenty-outlet design, the predictable bound is $q_n = 1/20$ conditional on the recorded past. For other designs, derive the conditional bound at the actual analysis unit using Section 14’s allocation, leakage, selection and stage-dependence assumptions. Use the comparison-family error budget specified in Section 14.

17. Research ethics and reproducibility

Research ethics

The proposed studies enrolling human participants require informed consent, freedom to withdraw and independent oversight. Protect privacy, minimize burden and preserve treatment and comfort (World Medical Association, 2024). Animal studies apply replacement, reduction and refinement principles and report methods and outcomes transparently (Percie du Sert et al., 2020). Prospective review of the mosquito experiments addresses species, conditions, numbers, endpoint procedure, containment and scientific justification.

Minimum reproducibility capsule

Archive the timestamped protocol, acceptance geometry, planned sensitivities and outputs with Section 13's machine-readable verdicts and recovery records. Document data provenance, transformations, linkage, time windows, splits, software, seeds and code. Record exclusions by stage and preserve Section 14's event and allocation logs with the raw recordings. Publish artifact hashes under the study's ethics, privacy and licensing plans. For restricted data, specify the access procedure or a documented substitute and the independent checks it supports. The Longitudinal Aging Study Amsterdam requires an approved proposal and data-sharing agreement.

Save PTVC_Companion.zip from the PDF attachments panel or open the separate copy in the publication package. It contains

code, reference outputs and checksums for three constructed benchmarks. The empirical supplement provides source manifests, replication scripts and sensitivity analyses for the three human datasets.

For a confirmatory study, an independent auditor checks the Section 13 decision rules and blinded recovery results against Sections 2–12 and 16 before unblinding, then checks the completed verdicts against those rules.

The next test of compensation

The next experiment can compare bounded episodes that reach the same rating with the same time left. When their accrued balances differ, does the compensation model predict the difference in signed area over the delayed window more accurately than each of its two equally informed rivals?

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Competing interests. Stephen Takowsky originated PTVC, founded the Law of Fairness Project, and owns or controls rights in related publications and project materials.
Data, code, and materials. See the Minimum reproducibility capsule.
Ethics statement. The empirical analyses use secondary data.
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About the Author

Stephen Takowsky has spent his life studying complex systems: how rules shape them, how people cooperate and compete within them, and how hidden constraints can change what becomes possible. His path has crossed finance, competitive strategy, professional esports, and the leadership of global online communities, ultimately leading to the Law of Fairness project, a decades-long effort to investigate whether fairness itself may reflect a deeper structure of conscious life.

Competing as "MrGameTheory," Takowsky reached number one on the Civilization Revolution leaderboard and in the international Civilization IV and Civilization V leagues while pursuing a career in finance at Morgan Stanley in Beverly Hills. In 2016, aXiomatic, an ownership group led by Peter Guber and Ted Leonsis and backed by investors including Magic Johnson, Steve Case, and Tony Robbins, acquired a controlling interest in Team Liquid. Following the acquisition, Team Liquid selected Takowsky to captain its newly created Civilization team, later recognized by Guinness World Records as the first Civilization esports team. Across thousands of games, Civilization became an unconventional education in strategy: understand the whole system, think far beyond the next move, anticipate intelligent opposition, and keep searching when the obvious paths fail.

Strategy then became community leadership. Takowsky spent seven years at 505 Games, serving as Senior Digital Content & Engagement Manager and leading global communities for Terraria, which has surpassed 70 million copies sold, as well as Ghostrunner, WUCHANG: Fallen Feathers, Eiyuden Chronicle: Hundred Heroes, and other titles. He also founded the United Servers of Discord Alliance, bringing together leaders from major communities built around Roblox, Pokémon, Star Wars, League of Legends, Grand Theft Auto, Warframe, Animal Crossing, and other properties. Earlier, he served as administrator and producer for TF2Outpost, a major Steam digital-goods marketplace. Across these very different environments, he kept returning

to the same insight: better structure can expand what people are capable of building together.

Throughout those years, one question kept following him: could the apparent unfairness of life conceal a deeper structure? That question became the Law of Fairness. Takowsky began developing the idea more than twenty years ago while studying Religious Studies and Philosophy/Law and Society at the University of California, Riverside, and gave it its first public form in his 2009 book, Of Grandeur. In the years that followed, he sought out academics and serious thinkers across many disciplines, asking for insights, criticism, and sources that could help separate fact from speculation. What began as a philosophical intuition gradually became a research program built to face mathematics, measurement, competing explanations, and evidence.

The ten-book Law of Fairness series follows that question outward, from consciousness and scientific testing into ordinary life, meaning, and civilization. If the Law is true, then there may be no "free happiness" in the sense of positive experience remaining unresolved by the lifetime balance the hypothesis predicts. That possibility raises a profound practical question: what if humanity has been pursuing the wrong model of utopia? Rather than organizing life around the endless pursuit of pleasure, the series explores whether greater self-moderation, less needless suffering, and better-designed institutions could help remove some of the forces that keep people and societies trapped in conflict, corruption, and instability.

Takowsky's fifty-year horizon is designed so that building a better future does not require tearing apart the present. It gives institutions and communities time to test reforms, build infrastructure, prepare future leaders, improve technology, and correct mistakes before they spread. Just as importantly, it gives today's generation the chance to be remembered for beginning something larger than itself: laying the

foundations for a more transparent, peaceful, and prosperous world it may never live to see.

The project's future is collaborative. Takowsky welcomes researchers, universities, laboratories, funders, technologists, students, critics, writers, artists, community builders, and volunteers willing to bring rigor, imagination, and courage to one of humanity's oldest questions. The next breakthrough may come from a theorem, an instrument, a dataset, a counterexample, or an experiment no one has yet imagined. The Law of Fairness began as one person's question. Its next chapter belongs to everyone willing to help discover what is true and build what that knowledge makes possible.

Explore the project and its official communities at LawOfFairness.com. Research, scholarly, media, funding, volunteer, and collaboration inquiries: team@lawoffairness.com.

